# Santiago's A.T. Field: Visualizing Urban Accessibility through an *Evangelion*-Inspired Interface

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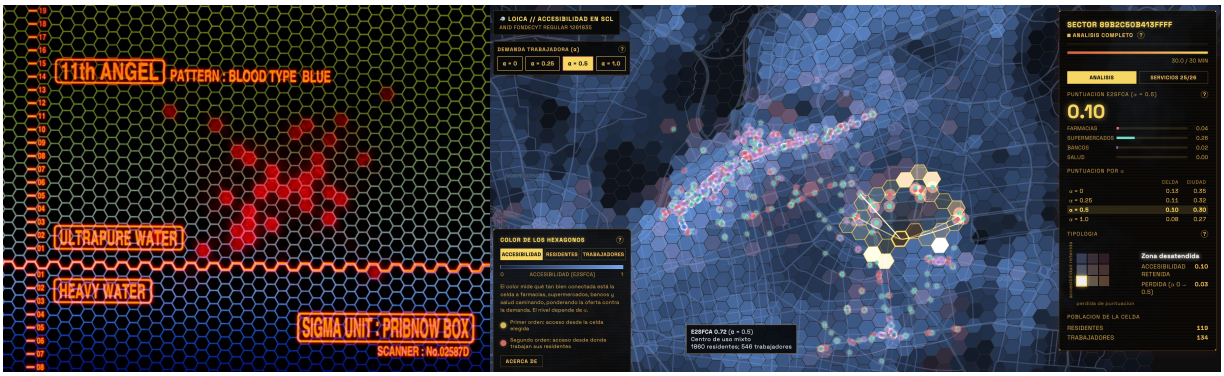



Figure 1: A fictional display and its translation into a working visualization. Left: a frame of Episode 13 of *Neon Genesis Evangelion* [1], with the interface used to identify the Angel Ireul. Right: Santiago's A.T. Field, an interactive visualization of urban accessibility.

## Abstract

Science-fiction interfaces are often reproduced for their appearance, and their diagnostic logic is rarely rebuilt around real data. Here we describe Santiago's A.T. Field, an interactive visualization of urban accessibility built from a diagnostic interface in Episode 13 of *Neon Genesis Evangelion*. We translated that interface in five stages: phenomenon, field, progression, interface, and response. A microscopic Angel became a relational urban problem, and a fictional scanning lattice became an accessibility field computed with the E2SFCA method. Every element of the fictional display is tied to a quantity of the model. Three tensions appeared when we made the fictional language operational: the saturated palette of the genre against color-vision deficiency, the motion of the genre against the reduced-motion preference, and the per-frame cost of the animation. The urgency of the fictional command center has its own cost, because emergency aesthetics can cast urban populations as threats. These three tensions are what the translation cost, and they are what transfers to other attempts of this kind.

**Index Terms:** Science fiction, urban accessibility, visual analytics, progressive visualization, mobility.

## 1 Preface

This paper answers SciFi-VIS's invitation to examine the two-way relation between science fiction and visualization [7] through one concrete act of translation. *Neon Genesis Evangelion* is a 1995 Japanese animated series in which the organization NERV uses dense diagnostic interfaces to interpret attacks by entities called Angels. Our visualization began with one of those instruments. We asked what had to be modified for it to operate on a real problem, through a process of *constructive translation*, which means rebuilding a fictional visualization around a new analytical object. The five sections that follow are the five stages of that work, and a last section states what it cost. The interactive system is available [10].

Research on science-fiction interfaces has surveyed how films depict interaction [15, 17], and it has described fictional devices as prototypes of real technology [9]. In visualization, the superpowers of fiction have been used to characterize and inspire systems [18]. Planning research has reviewed accessibility instruments, which are tools to compute and display accessibility for practitioners [13].

Studies of *Evangelion* have analyzed its visual style [14], but we found no study of its interfaces. The series belongs to a wider circulation of anime outside Japan. In Latin America, comic artists have appropriated manga aesthetics and adapted them to their local contexts [8]. In a similar way, we adopted the interface aesthetics of *Evangelion* and adapted them to an accessibility instrument for Santiago.

## 2 Phenomenon

> *"It's an Angel."*
> *—Ritsuko, Episode 13 [5]*

In Episode 13 of *Neon Genesis Evangelion*, technicians find what looks like corrosion on a wall inside NERV. It spreads and resists removal, and its distributed behavior becomes visible only through the visualization interface, where it grows across a hexagonal lattice. It is Ireul, the 11th Angel [1].

We translate this to urban accessibility in Santiago de Chile, which is also a distributed relation among people, services, and movement. It appears in one place and is produced by the whole system. The two objects also differ. Ireul is a hostile entity, and urban accessibility describes the opportunities and exclusions of real inhabitants.

## 3 Field

> *"The pattern analysis is blue."*
> *—Ritsuko, Episode 13 [5]*

Santiago is tessellated into 9,025 hexagonal cells of 0.09 km$^2$ with the H3 grid at resolution 9. The H3 grid is built from hexagons be-

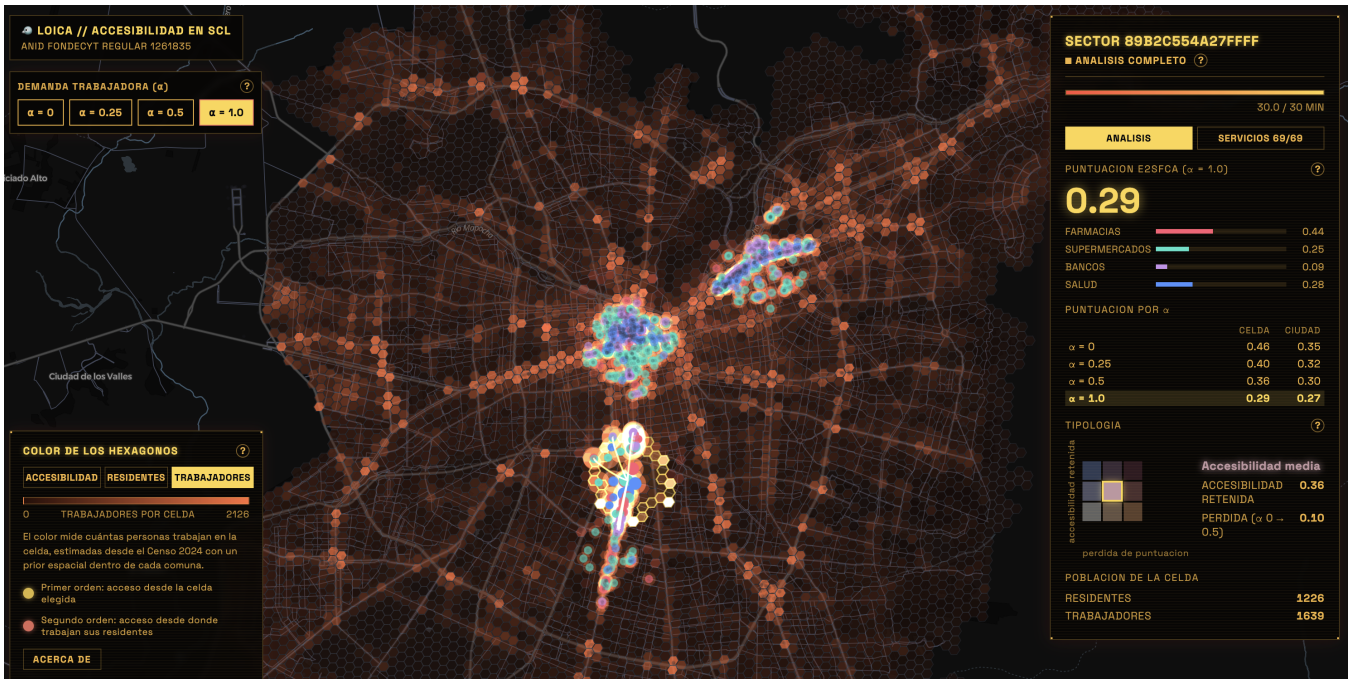



Figure 2: An intermediate state of Santiago's A.T. Field. The scan advances over the city and activates the services and the spatial relations that build the accessibility index.

cause the center of a cell is at the same distance from the centers of its six neighbors. Each cell receives an accessibility value. The prototype considers walking access to pharmacies, supermarkets, banks, and health services. Using the Enhanced Two-Step Floating Catchment Area method (E2SFCA), each service first receives a supply-to-demand ratio within a distance-weighted catchment; a cell's accessibility then accumulates the ratios of the services it can reach [11].

Our implementation extends this relation through the city's residence–work structure. The demand of a cell is its residents plus $\alpha$ times its workers, and the interface offers four values of $\alpha$: 0, 0.25, 0.5 and 1. With $\alpha = 0$ the index is the residential E2SFCA, and with $\alpha = 1$ a worker competes for the services around the workplace as much as a resident. The right value depends on how often workers use services near their workplace, and our data do not measure it. For this reason, $\alpha$ remains a parameter, and the interface opens at the midpoint, $\alpha = 0.5$. We call the result commuter-adjusted accessibility. Selecting a cell shows what its residents reach from home and what they reach from the places where they work, so the value of one hexagon depends on the rest of the city.

This computed surface is Santiago's A.T. Field. In *Evangelion*, A.T. stands for *Absolute Terror*, and the field is a barrier that blocks hostile contact. Here it stands for *Accesibilidad y Trabajo*, Spanish for accessibility and work. Both displays use the same hexagonal form (Figure 1). The fictional field blocks contact and this one measures it.

## 4 Progression

*"Evolution. . . "*
*—Fuyutsuki, Episode 13 [5]*

The interface scans the city, locates services, and activates the relations through which the index is assembled (Figure 2). Every element of the scan is tied to a quantity of the model. The front of the pulse advances over walking times computed with R5 [4] over the OpenStreetMap network, so its shape follows the street network. Walking times were computed at 3.6 km/h and end at 30 minutes. The catchment weight falls from 1 below 10 minutes to 0.03 above 20, so services beyond a 20-minute walk barely add to the index. Each service lights up when the front reaches it, and its size and brightness follow the weight of the E2SFCA catchment, so a service that adds little to the index is drawn small and dim. Gold arcs join the cell to the services reached from home, and red arcs join it to the services reached from the workplaces of its residents.

Progression exposes how services, catchments, and demand accumulate. The model describes cumulative reachability, and the front of the scan marks the walking time from the selected cell.

## 5 Interface

*"These are back doors!"*
*—Maya, Episode 13 [5]*

Both interfaces overlay a hexagonal lattice on a dark field, with cool context, warm active regions, and luminous technical labels. The result reads as an institutional diagnostic system.

In the fictional display, these elements need only sustain narrative plausibility. In the real interface, they must carry reproducible meanings. The grid encodes four variables and each one has its own ramp: blue for accessibility, gold for residents, orange for workers, and a bivariate three-by-three palette that crosses retained accessibility against the share of the daytime population which arrives to work.

The data were prepared with the following Python libraries: tobler for the interpolation of census counts to the grid, QuackOSM for the extraction of pharmacies, supermarkets and banks from OpenStreetMap, and r5py [6] for the walking times. The interface was implemented with React, deck.gl for the map layers, MapLibre GL for the basemap and h3-js for the cell geometry.

Three differences appeared when we made the fictional interfaces into a real one. First, the dark interface asks for saturated color, and the bivariate palette has to stay separable under color-vision deficiency. We built that palette in CIELCh space for a

printed figure, and we validated its nine opaque colors with a physiological simulation of the three deficiencies [12]. The interface lightens those colors for the dark basemap and raises their opacity with the value. The validation does not cover these modified colors. Second, the genre relies on motion, and some users set a reduced-motion preference in the browser. With that preference, the assembled field appears without the scan. A control in the legend also turns the scan on and off. Third, the animation has a frame budget: we animate inside the color accessor of each layer, because rebuilding layer data on every frame re-tessellates the grid and the frame rate drops.

The fictional display claims authority before its measures are explained. Our reconstruction keeps that authority, and the explanations and the parameters work against it, because they expose the assumptions of the model.

## 6 Response

> *"The end point of evolution is death."*
> *—Gendo, Episode 13 [5]*

Episode 13 stages visual analytics under emergency. An unfamiliar phenomenon changes faster than the institution can understand it. Automated systems detect its patterns but cannot respond alone, and Ritsuko interprets the adaptation and intervenes. Visualization mediates detection and action under time pressure.

Urban accessibility is a *slow emergency* that accumulates through ordinary journeys and unequal infrastructure. Santiago's A.T. Field imports the urgency of the fictional command center into this structural problem, and asks where connection is insufficient.

Emergency aesthetics can make persistent inequality immediate, and they can also cast urban populations as threats. We have not resolved this tension.

## 7 Outlook

The system covers Santiago, and it has not been used in planning practice yet. A replication in another city needs three inputs: resident counts, an OpenStreetMap street network and the location of jobs. The walking range could also become a control. This change requires precomputing the index for each range, in the same way as for the four values of $\alpha$.

The scan presents the index as a sequence, which relates it to narrative visualization [16]. Studies of charts show that visual embellishment and other visual features change what readers remember [2, 3]. A study could compare the scan with a static map of the same index, and measure what readers understand and remember from each one.

## 8 Transfer

Santiago's A.T. Field translates a chain of relations. The fictional scanning lattice became an analytical substrate, and the dramatic animation became an explanation of how the index is built. The three tensions of the Interface section are the cost of the translation, and they are what transfers to other attempts at making a science-fiction instrument operational.

*Evangelion* was released in 1995 and set its story in 2015. Santiago's A.T. Field carries one of its instruments into a working interface in 2026.


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